

P01 - Behavioral Finance

Empezar tarea

- Fecha de entrega 21 de sep a las 18:00
- Puntos 10
- Entregando una carga de archivo
- Disponible 9 de sep en 18:00 - 21 de sep en 18:00

Overview

In teams of 3, build a simulator that generates synthetic trading records, inject two behavioral biases at magnitudes you choose and record in advance, and then estimate those magnitudes back out using the standard estimators from the literature.

The point is not to discover that the disposition effect exists, we already know it does. The idea is that with real data you can never tell whether a positive finding reflects a real bias or a bug in your code because we have no ground truth (with a simulator we should have the ground truth). If your estimator can't recover a parameter you injected, then your estimator is wrong (and you should know it).

Simulator setup

Generate N traders, with $N \geq 1000$, where each trader has:

- δ_i : **disposition strength** 0 (none) to 1 (strong)
- κ_i : **overprecision / churn intensity** 0 - 1
- W_i : **starting capital** \$10,000 to \$500,000
- n_i : **number of positions held** 5 to 30

Make sure that δ_i and κ_i are drawn independently.

Prices

Simulate daily prices for a universe of at least 50 securities over at least 500 trading days. Geometric Brownian motion is good enough I guess (for a LAF...), but a factor structure with correlated returns is

better and of course will be rewarded (you'll have to explain it tho).

Big constraint: Again, future returns must be independent of any agent's decision to trade, if not, a high-turnover agent can buy securities that subsequently outperform (systematically), so this will create informed trading in your simulator (we don't want that, we'll end up with the wrong sign).

Trading costs

Every transaction has an explicit commission and a bid-ask spread. State both in USD or basis points. Without costs, overconfidence has nothing to detect.

Decision rules

On each day, each agent decides for each held position whether to sell. Do not parametrize PGR directly, writing `if position is a gain: sell with probability 0.15` and then estimating

$\widehat{\text{PGR}} = 0.15$ will just reflect your if statement, nothing to demonstrate. The injected parameter must sit one level below the quantity you estimate, like:

- A holding hazard rate that differs by domain: the immediate probability of selling is scaled by $(1 + \delta_i)$ when the position is above its purchase price and $(1 - \delta_i)$ when below
- A reference point utility where the agent sells when realized utility from closing, evaluated against the purchase price with a concave gain arm and convex loss arm, exceeds the continuation value

Either way, both PGR and PLR are emergent properties of the simulations, not inputs. Turnover likewise emerges from κ_i acting on trade frequency.

Scenarios

Run all eight. Each is a full population regenerated under the stated configuration:

#	Scenario	δ	κ	Confound active	Expected finding
1	null	0	0	none	Both estimators quiet
2	Disposition only, low	0.3	0	none	Disposition positive, overconfidence quiet
3	Disposition only,	0.8	0	none	Disposition positive (larger),

	high				overconfidence quiet
4	Turnover only, low	0	0.3	none	Disposition quiet, overconfidence negative
5	Turnover only, high	0	0.8	none	Disposition quiet, overconfidence more negative
6	Both active	0.8	0.8	none	Check for interaction of both estimators
7	Rebalancing confound	0	0	rebalancing	Disposition likely fires spuriously
8	Mean-reversion confound	0	0	belief in reversal	Disposition likely fires spuriously

Estimators

Disposition effect

On every day an account sells anything, classify each held position as a paper gain or a paper loss against its purchase price. Accumulate four counts across all accounts and all sale days:

- G_r : gains realized, G_p : gains still held
- L_r : gains realized, L_p : gains still held

$$\widehat{PGR} = \frac{G_r}{G_r + G_p}, \quad \widehat{PLR} = \frac{L_r}{L_r + L_p}$$

Report both the difference $\widehat{PGR} - \widehat{PLR}$ and the ratio.

Additionally, here are some things you must state explicitly:

- Days on which the account sold nothing contribute nothing to any count.
- Does selling half a position count as one realization, or as a fraction? (partial sales)
- Average cost, FIFO, or per-lot? these give different answers (cost basis)
- What happens to a position at exactly its purchase price?

And finally, bootstrap by **account**, not by transaction. Observations within an account are dependent, and treating them as independent will understate your standard errors by a large factor.

Overconfidence

For each account i , regress the performance turnover:

$$r_i = \alpha + \beta \text{Turnover}_i + \gamma' X_i + \epsilon_i$$

with X_i containing at minimum portfolio size, number of positions, and a risk exposure measure.

Run this twice, first on gross and then on net returns. The prediction is that $\beta \approx 0$ for gross and $\beta < 0$ for net, this contrast is the entire test:

- If only net returns fall with turnover, trading is expensive but not misinformed. This is the Barber and Odean (2000) result.
- If gross returns also fall with turnover, something specific is happening and you must identify what is going on.

On a negative gross slope, do not assume something is wrong with your code, and do not assume it is a finding. In real data it is a genuine result: Odean shows the securities investors bought underperformed the ones they sold by roughly 3.2 points over the following year, before costs. Trading there was misinformed, not merely expensive. There are 4 things that may produce that same slope, only the first one is an error:

- Violating the big constraint in prices: predictive information leaked into the trade decision, so agents systematically buy securities that go on to underperform, this is an error, fix it.
- Cash drag: If selling leaves an agent in cash before redeploying, high-turnover agents spend more time out of a positively-drifting market, no informed trading required.
- Spread inside the price path: if fills occur at bid and ask, your gross return is already net of half your transaction costs. A definitional error, not a behavioral one.
- Compounding path effects: even with i.i.d. returns, turnover changes the geometric return of a portfolio. This interacts directly with δ , since the disposition mechanism is a winner-selling rule.
- Reverse casualty: If your selling rule conditions on gains (and the disposition mechanism does by construction), then an account that happened to perform well is holding more winners, sells more of them, and books higher turnover. Turnover is then partly an outcome of the return rather than a cause of it, and the regression reads the arrow backwards. This one produces a positive slope, and it can be large enough to swamp the cost effect it is supposed to measure. Expect it in every scenario whose selling rule keys on the purchase price, and in any confound that trims winners.

Diagnose which one you have, if it's the first one, then don't report it, just fix it. The rest are legit so make sure to include a section detailing which one occurred.

Independence

δ and κ interact mechanically. An agent with a strong disposition effect holds losers indefinitely, which suppresses their trade count, which lowers their measured turnover. If you generate the two parameters with any correlation, Overconfidence's turnover variable is partly caused by Disposition's bias and the coefficient β becomes uninterpretable, so we must:

1. Draw δ_i and κ_i independently
2. Report the realized sample correlation between them
3. Report the correlation between δ_i and realized turnover, which will not be zero even when (1) holds, explain why
4. Discuss what would happen to your Overconfidence estimate if the parameters had been correlated by construction

Explaining (3) will require you to understand the difference between the injected parameter and the behavior it produces, so good luck.

Deliverables

- Written report documenting the following:
 - Simulator design with the injected mechanism
 - A results table covering all 8 scenarios showing the injected parameters, recovered estimates and standard errors
 - Analysis covering the relevant scenarios (only the ones where something interesting occurred), the boring ones may have just a short paragraph
 - What the estimators can't distinguish and what data would be needed to distinguish it
 - Conclusions
- Code:
 - Jupyter notebook is fine
 - Py files even better
 - Github repo way better
- Pre-analysis:
 - One paragraph before running any estimator stating the parameters you will inject and what

you expect each estimator to recover.

Assessment

- Simulator design **(20%)**
- Treatment of confounds (scenarios 7 and 8), and what the estimators can't separate **(30%)**
- Validation (null recovery, monotonicity, cross-estimator quiet), **(20%)**
- Standard errors and clustering **(10%)**
- Reproducibility **(10%)**
- Written clarity and concision (English or Spanish) **(10%)**

If you produce all eight scenarios where no insights are found, then you'll most likely fail this project, you either got an extremely lucky simulation or did not examine the data close enough.

References

Read them wisely, they are in the "Papers" section in canvas...

- Are Investors Reluctant to Realize Their Losses? Odean, T.
- Trading Is Hazardous to Your Wealth. Barber, B. and Odean, T.
- Do Investors Trade Too Much? Odean, T.
- What Drives the Disposition Effect? Barberis, N. and Xiong, W. **(I included a summary version of this paper)**
- Prospect Theory, Mental Accounting, and Momentum. Grinblatt, M. and Han, B.